

Video Transcript

Video 6: Interfacing with a Database Container (04:31)

Previously, we installed the database server of MySQL in our local environment, and we did so in the traditional way. That is, you navigate to a site, you download those files, you go through the installation, the configuration and so on, and then we access that server using a Python program. Now, later we did the exact same thing, but we did it using a container that is on top of our Docker platform, we pulled an image from the container registry and we created a new instance of MySQL, but now running in a container. So, let's go ahead and access that container that we have and see if in fact it is the same as accessing a server that was set up in the traditional way, downloading the files and going through the installation. As you can see, I have here my editor and I'm going to go ahead and drag and drop a folder that I have on my desktop and it has a single file inside of a folder called Sample.

I'm going to go ahead and click on it. And as you can see there, it's a pretty simple file, it's code we've used before. The very first block sets the driver that we need to use, then a couple of packages that we need for the code, one is for our timestamps and the other one is a unique identifier. The very first part of it establishes the connection to the database. And if you see, there are no differences to what we did before, it is exactly the same, except now we are running against that database that is inside of a container.

The rest of it is the same, this is the cursor that we create and then the SQL command that we're going to execute. And in this case, we're simply going to list the databases. Then we're going to, if we get some results, we're going to iterate through them and write the results to the console and we'll end up by closing that cursor and that connection. So, let's go ahead and open up the console here, the terminal. As you can see, this happens to be in my desktop, and inside of it I have that single file.

And now I'm going to go ahead and enter `python3 show databases`. And if I execute that command, you can see there that I get the default databases that are part of that installation. Now, as you can see here, I have opened the docker dashboard. You can see there my container, MySQL which I have named some MySQL and below it I am in the same location that I was inside of the editor that is on my desktop inside a folder called Sample and inside of it I have 'show

databases.' Now, if I enter the same command line instruction that I did before, you can see here that once again I get the databases that we saw before.

Now, however, if I go here and I stop the container and in a moment the color should change there from green to gray and I run this again, you can see there that the container is no longer available. And if I restart it, let me go ahead, and this is already up here. It took a moment and I clicked it twice. Then you can see there that the databases are available once more. So, you can see here that we're running inside of a container environment. You can see here as well that if we simply delete this, then it's gone and you will no longer have access to it.

Everything that you had inside of that container has now been erased. Now, this is quite convenient when it comes to being able to create and iterate quickly through different scenarios and configurations and certainly to be able to create databases at a high speed. So, go ahead and try this yourself. Make sure that you can access that container, make sure that you can stop the container and be able to remove it and recreate it once again.

